AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 20 · STUDENT LESSON

Image tensors and channels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Name the axes. Track the pixel. Check the values.

Convert image layouts while preserving channel meaning and distinguishing scaling from information loss.

Keep one habit: Complete RGB pixels → labelled channel-plane diagram → known-value round trip → pictured batch axis → scaling audit → independent new image.

MAIN LESSON · SECTION 1

Can you find the same colour measurement after the axes move?

> Name row,column,channel and batch positions before reading values.

Learning goals

I can read a pixel using declared row column and **channel** axes.

I can convert **channel** layout without changing pixel meaning.

I can distinguish fixed **value scaling** from information losing transformations.

Retrieve preserved and lost information

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why does a representation with two components not necessarily preserve two original measurements?

> **My starting explanation**

End goal

By the end, I can explain image **tensor**s and **channel**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Read a pixel using declared row column and **channel** axes

Convert **channel** layout without changing pixel meaning

Distinguish fixed **value scaling** from information losing transformations

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

One pixel contains three ordered colour values

> This image has four pixels and twelve stored channel values.

Complete RGB/zero-based axis contract

A small fictional RGB image has height 2 pixels, width 2 pixels and three **channel**s ordered red, green, blue. Store it as image[row][column][**channel**], abbreviated HWC.

Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are **channel**s 0, 1 and 2. Each stored component is an integer from 0 to 255.

Every pixel and colour triplet

Position	RGB triplet	Meaning
Row 0 column 0	[255, 0, 0]	Red

Position	RGB triplet	Meaning
Row 0 column 1	[0, 255, 0]	Green
Row 1 column 0	[0, 0, 255]	Blue
Row 1 column 1	[255, 255, 255]	White

Shape,channel meaning and other formats

The shape is 2 by 2 by 3. There are four pixels and twelve **channel** values. The blue component of the bottom-left pixel is image[1][0][2] = 255.

A single **channel** is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three **channel**s here; other formats can include additional **channel**s.

Declare the format instead of assuming every three-dimensional array means the same thing.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace pixels into planes and back

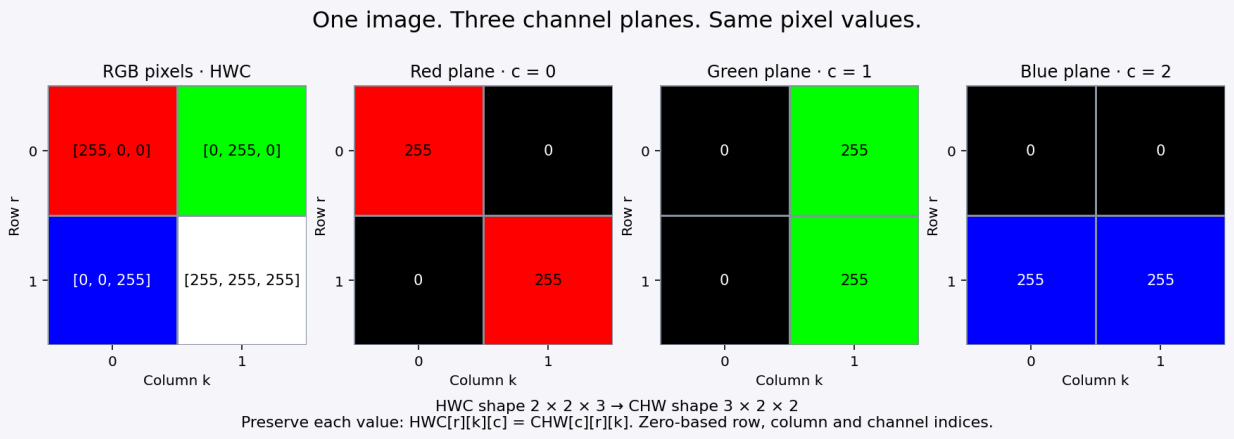
> A preserved value needs its row,column and channel labels in both layouts.

Complete axis-conversion definition

A **channel**-first layout uses image[**channel**][row][column], abbreviated CHW. Our image then has shape 3 by 2 by 2. A correct conversion preserves the rule chw[c][r][k] = hwc[r][k][c].

The code assumes a nonempty rectangular image with exactly three values per pixel.

Labelled RGB image and all three planes



Complete numeric planes and same-value indices

CHW channel index / name	Row0 values(column 0 then1)	Row1 values(column 0 then1)
0 /red	[255, 0]	[0, 255]
1 /green	[0, 255]	[0, 255]
2 /blue	[0, 0]	[255, 255]

Same measurement	HWC index	CHW index	Preserved value
Top-right green	[0][1][1]	[1][0][1]	255
Bottom-left blue	[1][0][2]	[2][1][0]	255

Same measurement	HWC index	CHW index	Preserved value
Bottom-right red	[1][1][0]	[0][1][1]	255
Top-left blue	[0][0][2]	[2][0][0]	0

Conversion versus regrouping and input assumptions

The complete converter assumes a nonempty rectangular image with exactly three valid raw channel values per pixel. It checks a known measurement and round-trip reconstruction for the supplied image, not every possible malformed file. CHW conversion reads the same value with a different axis order. Merely reshaping consecutive flattened values gives a wrong red plane even when dimensions look right.

Keep the RGB channel names, raw 0–255 range and traversal order with the shape.

Reading nested values in order and merely regrouping them into a new shape does not perform this axis conversion. The first four flattened HWC values are [255, 0, 0, 0].

A naive regrouping makes these the first **channel** plane, but the correct red plane is [[255, 0], [0, 255]]. Shape agreement alone cannot detect this mistake.

Complete runnable round-trip conversion

```
hwc = [
    [[255, 0, 0], [0, 255, 0]],
    [[0, 0, 255], [255, 255, 255]],
]
height, width = len(hwc), len(hwc[0])
chw = [[[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]
for channel in chw:
    print(channel)
assert chw[2][1][0] == hwc[1][0][2] == 255
restored = [[[chw[c][r][k] for c in range(3)]
             for k in range(width)] for r in range(height)]
assert restored == hwc
```

Connect each visual value to its named index

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change layout without changing pixel meaning

> A batch axis counts images;it does not add a colour channel.

Current flatten-versus-convert task

- A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured three-image batch

- Change: Add a batch containing three identical images in NHWC layout.

▼ **Check after predicting**

Prediction: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Track one named pixel through a layout and batch change

> Keep the pictured height,width and batch count distinct.

Complete separate pixel facts and scope

Move colour values between **tensor** layouts without changing the pixels.

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate two-row, one-column RGB image:top[1, 4, 7], bottom[2, 5, 8]. HWC shape 2×1×3 becomes CHW3×2×1, with red[[1], [2]], green[[4], [5]], blue[[7], [8]]. Bottom green stays 5at HWC[1][0][1]and CHW[1][1][0]. Three such images in NHWC have shape 3×2×1×3; bottom green in the first image is batch[0][1][0][1].

These small values are raw classroom examples, not the main red/green/blue/white image's255 values.

Trace the two layouts and bottom green

AI ACADEMY / AI-5/20 · 02 · TRACE THE CASE

Follow the deciding steps

01 Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.

02 Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

03 Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record HWC shape	Shape is [2, 1, 3]; top green is 4 and bottom green 5.
2. Reorder to CHW	Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].
3. Check same pixel	Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Predict the added three-image batch

- Change one thing: Add a batch containing three identical images in NHWC layout. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/20 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Changed condition

Add a batch containing three identical images in NHWC layout.

Worked changed-case result

Batch shape is [3, 2, 1, 3].
Why: The new first dimension counts images; it does not add a fourth colour channel.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain channel/schema limits and the batched index

- Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

- Read the labelled picture: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reconstruct the main image’s right-hand pixels

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image tensors and channels

The facts we will use

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/20

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.

2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Separate reordering,scaling and information loss

> A fixed division does not estimate a training statistic;averaging colours changes the information.

Complete scaling,double-scaling and colour-mean explanation

For this lesson, divide every **channel** value by 255 to obtain floating-point values from 0 to 1. Thus [255, 128, 0] becomes [1, 128/255, 0], with the middle value approximately 0.502.

This fixed conversion does not fit a statistic from data. Apply the same conversion during **training** and later use, and record whether an **input** is already scaled.

Dividing twice makes a raw value of 255 become 1/255, approximately 0.00392, instead of 1. Integer rounding after scaling can also erase distinctions.

In exact arithmetic the declared division is reversible by multiplication; finite precision or deliberate rounding can limit exact recovery.

Averaging RGB **channel**s is a different operation. The classroom mean gives red [255, 0, 0] and green [0, 255, 0] the same value 85, losing their colour distinction.

It is a simple illustration, not a perceptual brightness formula. If pre**process**ing instead estimates means or spreads from a dataset, fit those statistics using **training** data only.

Every transformation and its consequence

Complete transformation	Example result	Preserved or lost distinction
HWC→CHW	Same twelve values in declared new positions	Preserves every pixel/channel under correct conversion
One fixed division by255	[255, 128, 0]→[1, 128/255, 0]	Exact mathematical scaling is reversible; keep precision and raw/scaled status
Accidental second division	Original255→1→1/255≈.00392	Wrong pipeline scale despite unchanged shape
Simple channel mean	Red[255, 0, 0]andgreen[0, 255, 0]both 85	Colour distinction lost; not a perceptual brightness formula

Complete transformation	Example result	Preserved or lost distinction
Rounding scaled channel values to integers	128/255 can collapse to a rounded endpoint	Deliberately loses numerical distinctions

Try exact fractional scaling and the colour collision

- Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise axis names and value checks with less help

> State both the index and the measurement it identifies.

Current task and response

- For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

> **Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “Image **tensor**s and **channel**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured batch-index reasoning

- Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent new image

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Audit an axis conversion

Check your reasoning

Explain the picture

Run a complete pixel,axis and value-scale audit

> All pixels,planes,known indices and transformation probes are supplied.

Current wrong-plane task and complete converter

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

```
hwc = [
    [[255, 0, 0], [0, 255, 0]],
    [[0, 0, 255], [255, 255, 255]],
]
height, width = len(hwc), len(hwc[0])
chw = [[[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]
for channel in chw:
    print(channel)
assert chw[2][1][0] == hwc[1][0][2] == 255
restored = [[[chw[c][r][k] for c in range(3)]
             for k in range(width)] for r in range(height)]
assert restored == hwc
```

Complete runnable scaling/colour audit

```
# Fixed numeric preprocessing examples, not learned normalization statistics.
raw = [255,128,0]
scaled = [v/255 for v in raw]
twice = [v/255 for v in scaled]
print('raw RGB', raw)
print('scaled once', [round(v,6) for v in scaled])
print('scaled twice', [round(v,6) for v in twice])
print('undo one division', [round(v*255,6) for v in scaled])
red, green = [255,0,0], [0,255,0]
print('raw red / green', red, green)
print('simple channel means', sum(red)/3, sum(green)/3)
print('red with RGB values reversed', red[::-1])
print('boundary scaling', 0/255,255/255)
```

```
# Raw values are supplied valid RGB integers. This code does not parse or validate a file.
# The simple mean is not a perceptual luminance formula.
```

Normal,boundary,malformed,missing-schema and value checks

Complete tensor/preprocessing probe	Expected answer / action
Normal main HWC/CHW round trip	2×2×3→3×2×2→sameoriginalimage
Top-left red test	Raw RGB[255, 0, 0], not swapped[0, 0, 255]
Zero/full channel boundaries	0/255=0, 255/255=1
Flatten first four HWC values	[255, 0, 0, 0]regroups towrongred[[255, 0], [0, 0]]; correct[[255, 0], [0, 255]]
Ragged row, missing channel or unknown RGB/BGR order	Outside declared rectangular RGB contract; inspect/reject/request schema before conversion
Five main images stacked NHWC	5×2×2×3=60 values; NCHW5×3×2×2
Already scaled full-red value 1 dividedagain	1/255≈.003922, not expected 1
Learned channel mean/spread instead of fixed division	Fit those statistics on training data only; record a different preprocessing rule
Correct shape alone	Still check channel names, a known pixel, value range, scale count and reversible layout

Predict,convert,reconstruct and inspect actual values

1. Predict every plane and the two right-hand pixels before running or tracing the complete HWC/CHW program. Verify the full round trip and a known red/green/blue coordinate. Then use the complete scaling audit to compare one division, two divisions, RGB reversal and colour averaging. Keep raw/scaled status and original data with the record.

   The code assumes valid supplied arrays; it does not inspect arbitrary image files or validate missing channels. A shape match is one check, not proof of correct colour or preprocessing.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: convert a different tiny image and batch

> Use its height,width,pixel values and copy count.

Complete new independent image and task

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Convert a new image independently

Attempt this without hints

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save every axis name and preprocessing convention

> The project record should let a reader track the same pixel through the pipeline.

Project artifact and evidence table

Add a labelled image **tensor**s and **channel**s **evidence** sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and **channel** axes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one preserved pixel and one information-losing operation

> Explain why axes,channels and value conventions belong together.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Read a pixel using declared row column and **channel** axes

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Tensor	A numerical array with a declared set of axes
Channel	One component plane of an image such as red green or blue
Axis order	The sequence of meanings assigned to array dimensions
Value scaling	A declared numerical conversion applied to stored values

Use two terms to explain a step in this week’s example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.
- Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?
- The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent one-row two-pixel image and four-image batch for the first individual attempt.

Current explanation and response

- Stacking five same-size HWC images adds a leading example axis N. This batch has shape 5 by 2 by 2 by 3 and contains 60 **channel** values. With **channel**-first images, its shape is 5 by 3 by 2 by 2.

  The example axis is not another colour **channel**. Every axis must keep its meaning through the model.

- Use a tiny known image, named axis meanings and expected **channel** values to check a pipeline. Keep original images when testing transformations.

  Flattening can be reversed if shape and traversal order are retained; averaging colours generally cannot. Next week reads small neighbourhoods with a convolution operation.

- The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

> **A loader returns the expected shape, but red and blue channels were exchanged and already scaled values were divided by 255 again. Does the shape check establish correctness? Name an exact pixel check and a value check that expose these faults.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Move colour values between **tensor** layouts without changing the pixels.

By the end, I can	How I will show it
Read a pixel using declared row column and channel axes	Explain the deciding step, show a trace or test, and state one limit.
Convert channel layout without changing pixel meaning	Explain the deciding step, show a trace or test, and state one limit.
Distinguish fixed value scaling from information losing transformations	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Move colour values between **tensor** layouts without changing the pixels.
- Which labelled step matches this detail? Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.
- 1. Record HWC shape
- 2. Reorder to CHW
- 3. Check same pixel
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check same pixel Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. A **tensor** is an array with named dimensions.

HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices.

Adding a batch dimension counts images and does not add a new colour **channel**.

Changed-case question and optional hint

- Changed-case practice: Add a batch containing three identical images in NHWC layout. Which conclusion is supported by the supplied facts?
- 1. Batch shape is [3, 2, 1, 3].
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Repair and retain the record

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

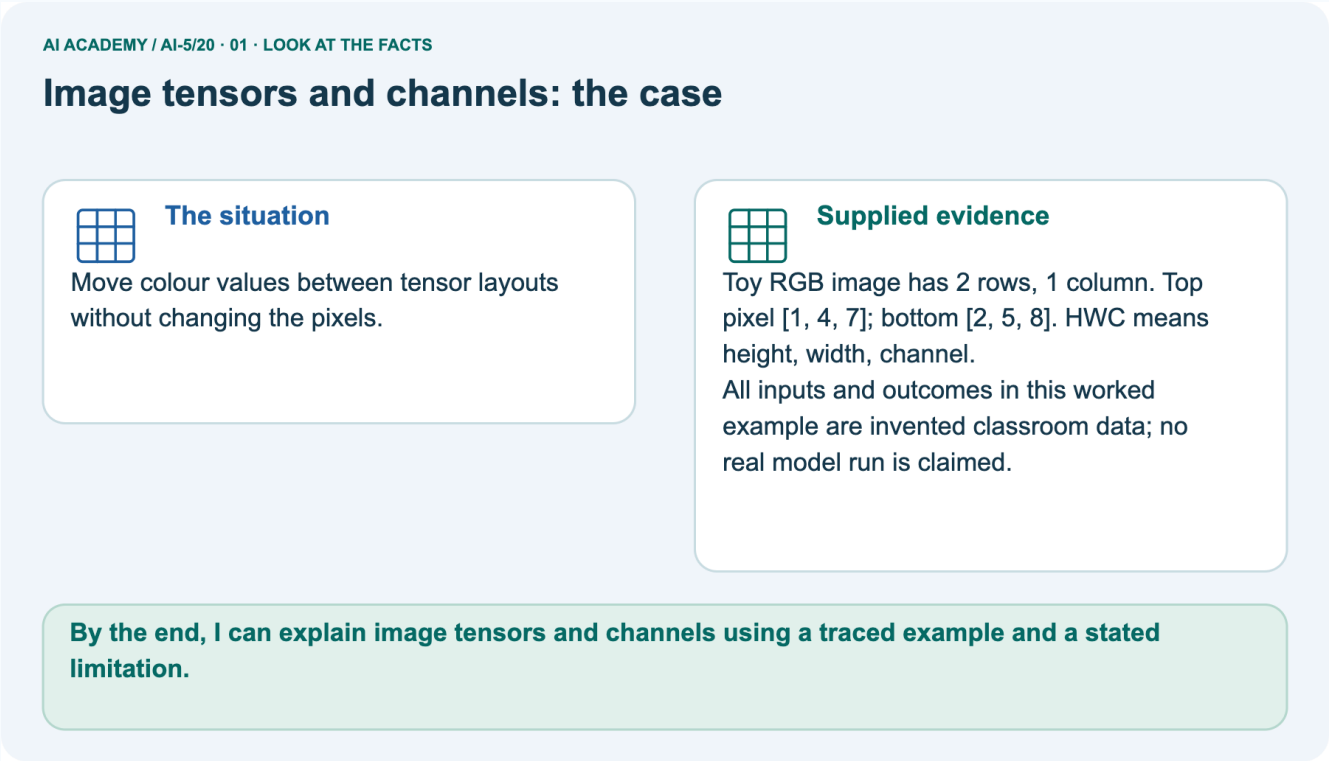
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Image tensors and channels: the case



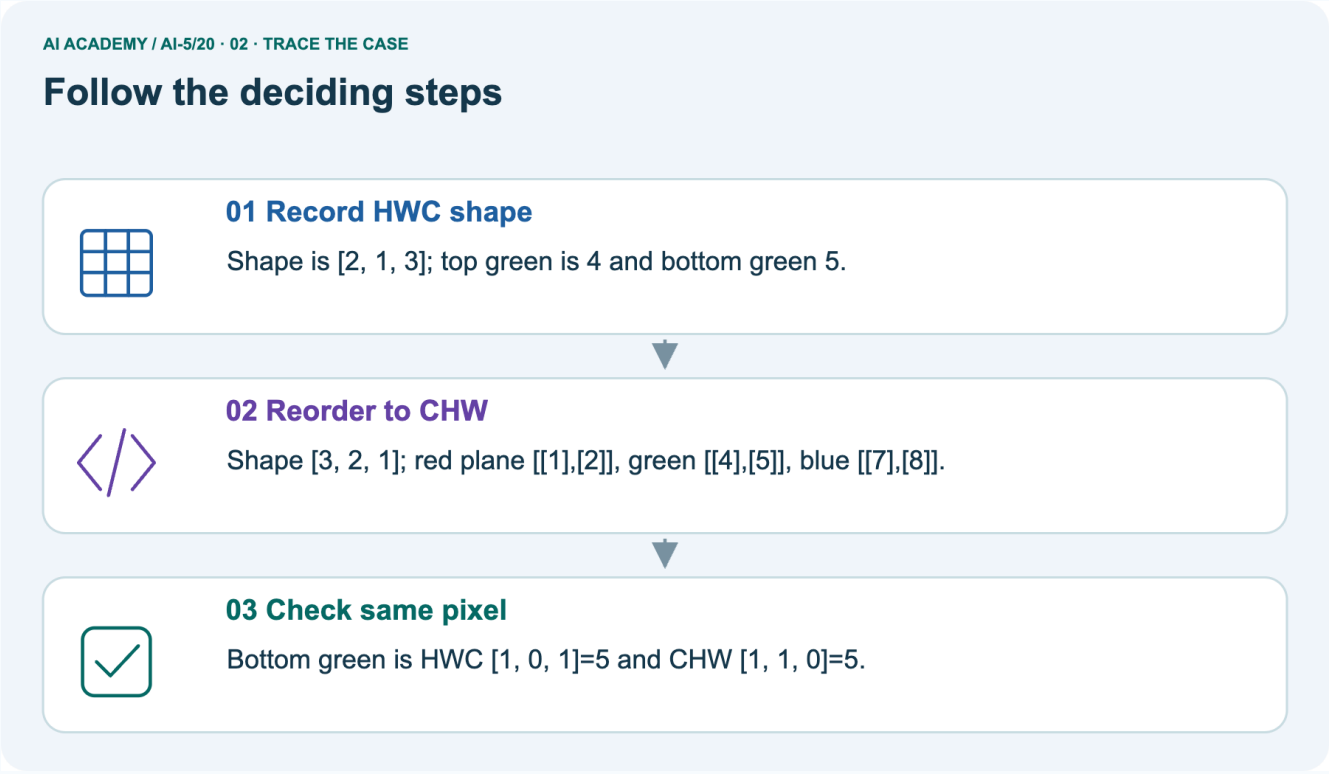
Follow the labelled picture with the matching explanation. Open the vector diagram

Move colour values between **tensor** layouts without changing the pixels. Supplied facts: Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

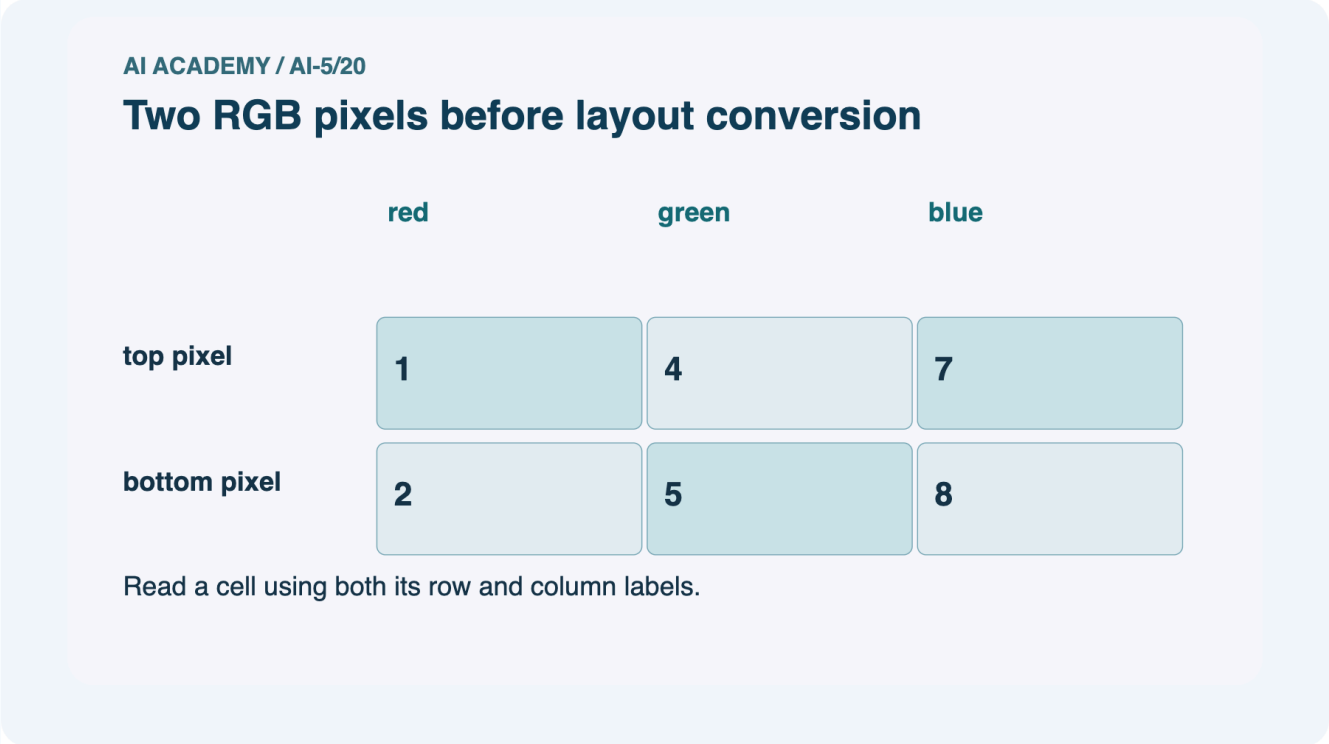


Follow the labelled picture with the matching explanation. Open the vector diagram

Record HWC shape: Shape is [2, 1, 3]; top green is 4 and bottom green 5. → Reorder to CHW: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]]. → Check same pixel: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two RGB pixels before layout conversion

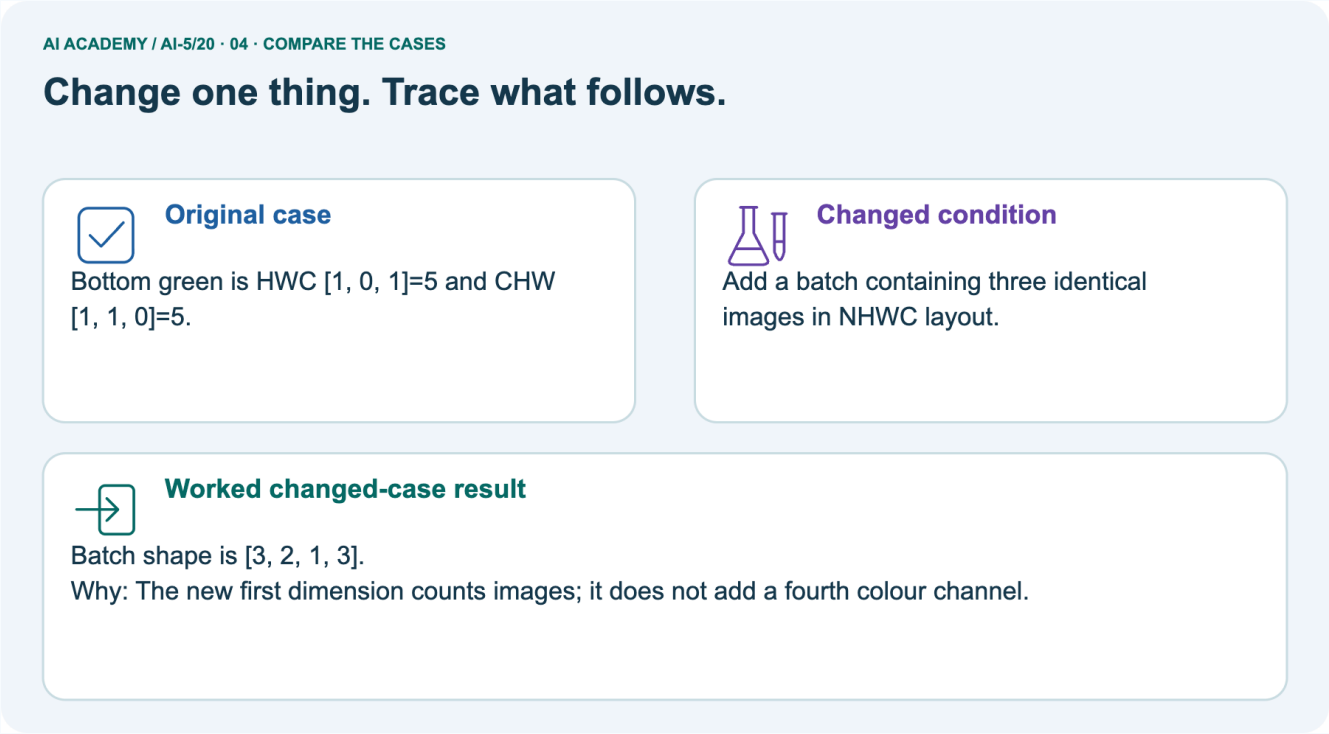


Follow the labelled picture with the matching explanation. Open the vector diagram

Two RGB pixels before layout conversion. top pixel: red = 1, green = 4, blue = 7. bottom pixel: red = 2, green = 5, blue = 8.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



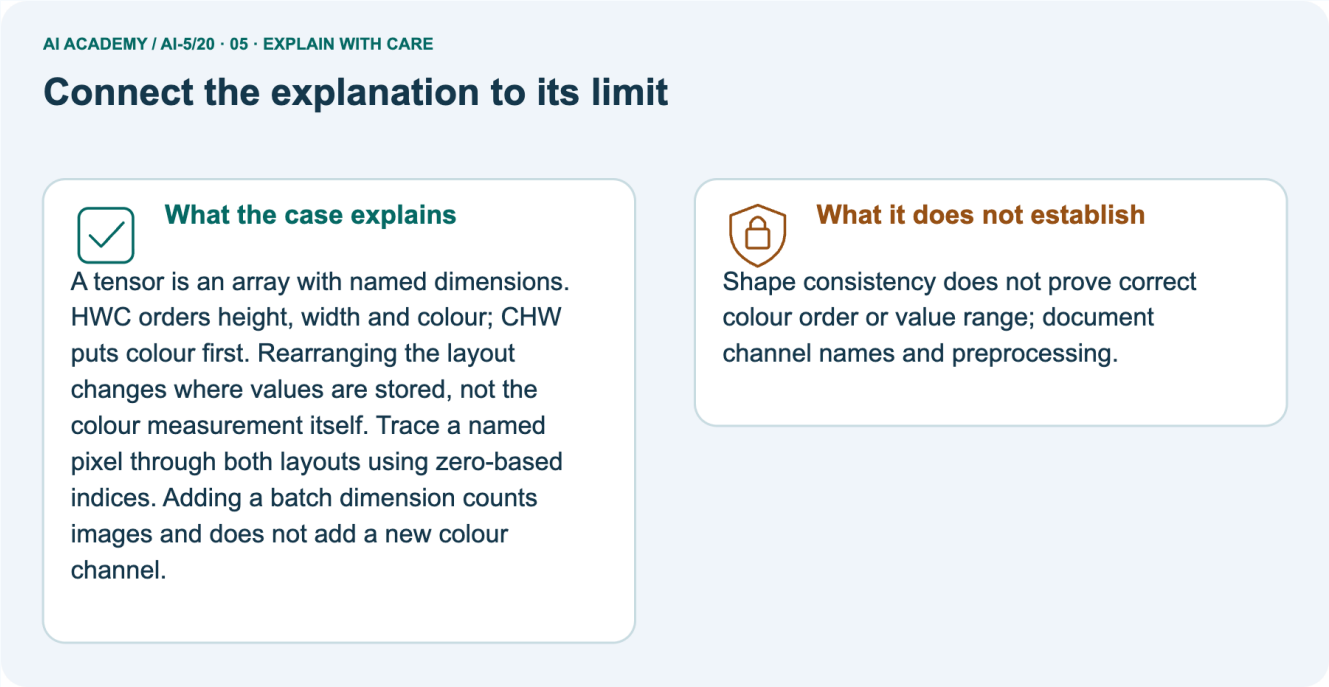
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. Changed condition: Add a batch containing three identical images in NHWC layout. Expected result: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

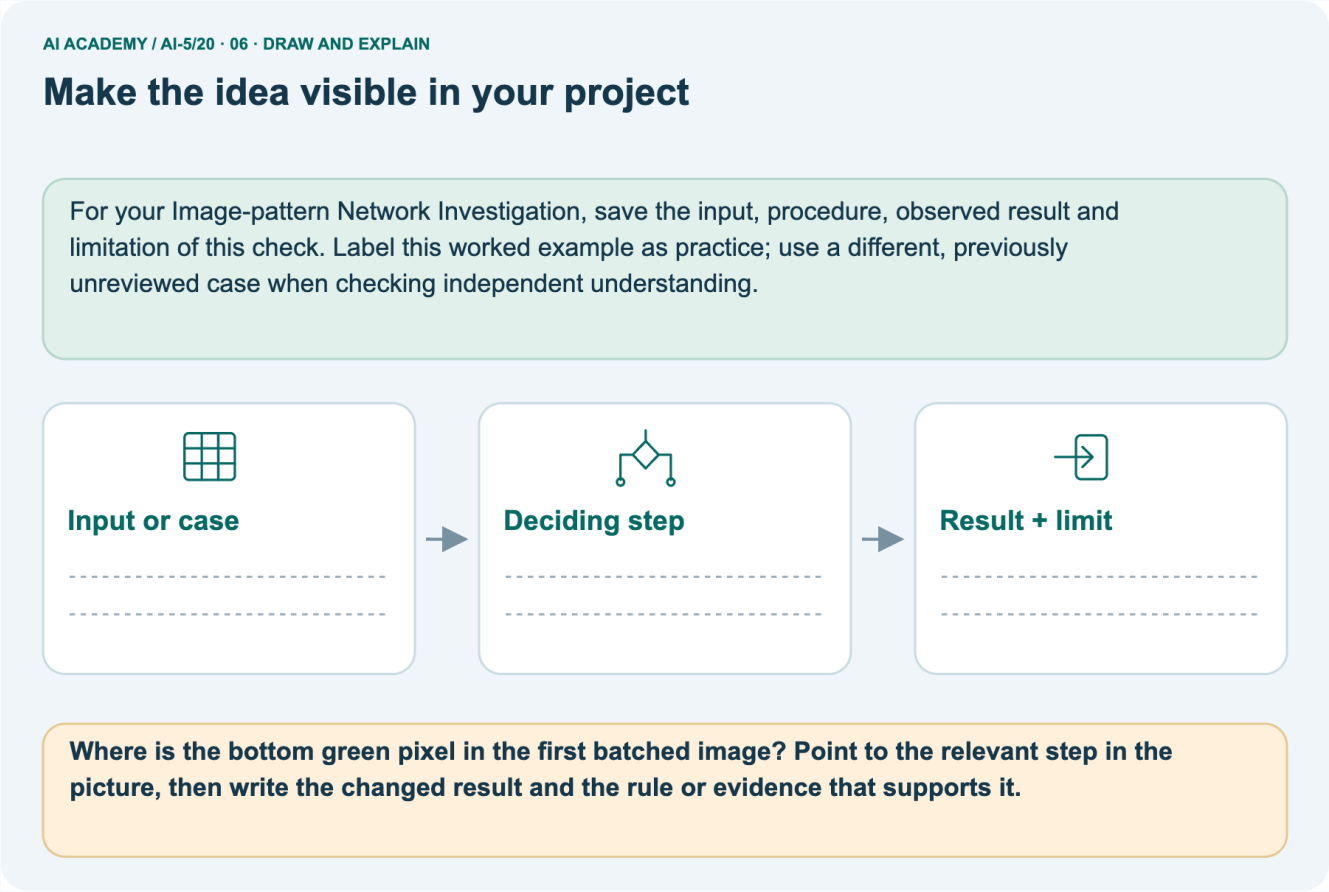
A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Limit: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Image tensors and channels: the case

02. Follow the deciding steps

03. Two RGB pixels before layout conversion

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.